REIHAN APRIANDI

Forestry & Spatial Data Science - Remote Sensing - Environmental Analytics
Pekanbaru, Riau, Indonesia | +62 888-0402-2931 | apriandireihan@gmail.com | linkedin.com/in/reihan-apriandi-700845266
github.com/Hansen-arch | hansen-arch.github.io

PROFILE

Final-year Forestry undergraduate with field and professional experience in land compliance supervision, above-ground biomass surveys, land cover analysis, and carbon project support. Integrates forestry expertise with Spatial Data Science, applying GIS, remote sensing, and Python-based analytics to evidence-based conservation, sustainable forest management, and climate-related research. Recent independent work analyses satellite observation coverage for flood monitoring across European and Indonesian river basins.

EDUCATION

Bachelor of Science in Forestry Aug 2022 - Present
Universitas Riau, Faculty of Agriculture - 8th Semester - GPA: 3.68 / 4.00

PROJECTS

Flood Coverage - Satellite Observation Coverage for Flood Monitoring Nov 2026
Solo project - Python, STAC, Sentinel-1, GloFAS - Public source

- Measured whether a satellite actually observes a flood while it is rising, using one year of real Sentinel-1 acquisition metadata across four river basins in Italy, Germany, and Indonesia (694 overpasses, 2025-11-01 to 2026-11-01).
- Computed empirical coverage instead of relying on nominal revisit: only 3 to 21 percent of 6-hour rise windows were observed, and Jakarta forecast peaks were checkable within a day 28 percent of the time against 91 percent on the Po.
- Built a dependency-free Python pipeline over the STAC and Open-Meteo APIs, with an interactive web front end and exportable figures.

BioSift - Species Occurrence Data Quality App 2026
Solo project - Python, Streamlit, pygbif, scikit-learn - Submitted to the 2026 GBIF Ebbe Nielsen Challenge

- Built a Streamlit web app that fetches GBIF occurrence records and runs ten automated quality checks, scoring each dataset from 0 to 100 for reliability.
- Added DBSCAN spatial outlier detection, kernel density previews, multi-species comparison, and export to Darwin Core Archive.
- Deployed publicly with the source code released, and documented the method for each check.

PROFESSIONAL EXPERIENCE

Site Compliance Supervisor Aug 2026 - Sep 2026
Ministry of Agriculture of the Republic of Indonesia (Kementerian Pertanian RI) - Contract - Teluk Meranti, Riau, Indonesia - On-site

- Monitored excavation activities on-site to ensure work stayed within designated plot boundaries and did not encroach on neighboring areas.
- Directed field workers on the placement of boundary stakes (patok) to mark excavation limits for each plot before work began.
- Guided excavator operators using the placed boundary stakes to accurately excavate irrigation channels, embankments (tanggul), and paddy field areas.
- Coordinated stake placement across multiple fields to clearly define plot boundaries and prevent excavation errors.

Assistant Surveyor - Above-Ground Biomass (AGB) Peatland Study Jun 2026 - Jul 2026
PT Hatfield Indonesia - Freelance - Riau, Indonesia

- Supported above-ground biomass (AGB) field surveys across tropical peatland areas as part of a carbon stock assessment project.
- Conducted forest plot measurements, including tree diameter (DBH), height, and species identification, to support biomass estimation.
- Recorded and managed field data using standardized data collection protocols to ensure data quality and consistency.
- Worked alongside senior surveyors to apply allometric equations and support carbon stock calculation workflows.

Assistant Analyst (Intern) Jan 2025 - Jun 2025
RECOFTC - The Center for People and Forests - Indonesia - Hybrid

- Digitized land cover and land use features from satellite imagery using predefined classification protocols with Collect Earth Online, ArcGIS, and QGIS.
- Verified land cover and land use datasets with approximately 90% accuracy, validated against field reference data using confusion matrix evaluation.

- Conducted field validation to cross-check satellite imagery interpretations against ground conditions.
- Coordinated with the project team to maintain methodological consistency across remote sensing and field validation workflows.

Capstone Project Participant - Student Stress Analysis Oct 2025

Hacktiv8 × IBM SkillsBuild - Online

- Used Python (Pandas, Matplotlib) for data cleaning, exploratory data analysis (EDA), and visualization.
- Produced a visual dashboard report identifying key stress drivers across subject types and learning environments, presented to non-technical stakeholders.

Forest Management Field Training Participant Jul 2024

HPGW, Institut Pertanian Bogor (IPB University) - West Java, Indonesia

- Completed field practicum covering community forestry, forest inventory, nursery management, soil conservation, non-timber forest products, block area mapping, and in-situ conservation methods.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Expert Staff, Commission 4 Jan 2024 - Dec 2024

Student Representative Council (DPM), Universitas Riau

- Contributed to institutional policy discussions and student governance within the campus legislative body.

Expert Staff, Fundraising & Business Division May 2023 - Sep 2024

Forestry Student Club (FSC), Universitas Riau

- Coordinated fundraising activities and managed business initiatives to support club operations and academic programs.

SKILLS

- **GIS & Remote Sensing:** ArcGIS, QGIS (Advanced), Google Earth Engine, GeoDa, GRASS GIS, Collect Earth Online
- **Land & Site Compliance:** Boundary Verification, Site Supervision, Encroachment Prevention, Permit Compliance
- **Accuracy Assessment:** Confusion Matrix, Target Accuracy of 90% or higher
- **Programming (Intermediate):** Python (Pandas, Matplotlib), R / RStudio, SQL
- **Web Mapping & Dashboard:** Leaflet.js, Folium, Streamlit
- **Machine Learning (Basic):** Supervised Classification, scikit-learn
- **UAV / Drone Data Processing (Basic):** Agisoft Metashape, OpenDroneMap
- **Other:** Exploratory Data Analysis, Data Visualization, Microsoft Office Suite (Intermediate)
- **Core Competencies:** Analytical Thinking, Attention to Detail, Field Research & Ground-Truth Validation, Above-Ground Biomass & Carbon Stock Survey, Cross-Functional Team Coordination, Time Management, Technical Report Writing
- **Language:** English Proficiency, B1 Intermediate

CERTIFICATIONS

- **Remote Sensing & GIS:** Advanced QGIS (Spatial Thoughts, Apr-May 2026); Visualizing Land Cover and Land Use Change with NASA Satellite Imagery (NASA ARSET, Feb 2026); Monitoring Water Quality in Lakes and Coastal Regions Using STREAM (NASA ARSET, Feb 2026); Hyperspectral Data for Land and Coastal Systems (NASA ARSET, Jan 2026); Advanced NASA Earth Observations and Tools for Active Fire, Smoke, and Post-Fire Monitoring (NASA ARSET, Jan 2026); Fundamentals of Remote Sensing (NASA ARSET, Dec 2025); GIS for Climate Action, 6 weeks (Esri, Mar 2026)
- **Environmental & Marine Data:** Marine Data 4 Sustainable Oceans Workshop (Copernicus Marine Service / Mercator Ocean International, Apr 2025); Marine Data 4 Asia Workshop (Copernicus Marine Service / Mercator Ocean International, Jun 2025)
- **Data Analytics & Artificial Intelligence:** Google Data Analytics (Coursera / Google, Oct 2025); Oracle Cloud Infrastructure 2025 Certified Data Science Professional (Oracle University, Oct 2025); AI Engineer for Data Scientists Associate (DataCamp, Aug 2025); Certified Data Analyst Associate (DataCamp, Aug 2025); Time Series Analysis (openHPI, Sep 2025)
- **Forestry & Environment:** Policy Analysis and Forest Governance 101 (RECOFTC, Jun 2025); Population Analysis Series: Tiger Density Analysis (Harimau Kita, Sep 2024); Advancing on REDD+ E-Academy (UNITAR / UN Environment, Oct 2024)